

MICHAEL HAYES

Senior VMware Engineer | VMware Cloud Foundation Specialist

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SUMMARY

Senior VMware Engineer with 25 years of hands-on experience architecting, deploying, and optimizing enterprise virtualization platforms. **Certified VMware Cloud Foundation 9 Administrator** with deep expertise across the full VCF 9 stack -- SDDC Manager, NSX, vSAN (ESA and OSA), VCF Operations, and VCF Operations for Logs -- spanning **deployment, integration, validation, and root-cause troubleshooting**. Recent work includes brownfield VCF recovery, log-level root-cause analysis and RCA authoring, 9.x lifecycle and 9.0-to-9.1 upgrade planning, and custom VCF Operations 9 monitoring integrations (management packs against storage REST APIs / NetApp ONTAP with dashboards, super metrics, and alerting), delivered as VMware by Broadcom solutions. Also spans Infrastructure-as-Code automation with VMware Aria Automation and REST API development (FastAPI / OpenAPI). Proven ability to modernize infrastructure, automate complex workflows, and deliver high-availability, high-performance environments at scale. Skilled in PowerShell, PowerCLI, Python, Bash, and SQL with a strong focus on automation, performance tuning, capacity planning, and cross-platform integration.

TECHNICAL SKILLS

- **VMware Virtualization:** GSX, ESX/ESXi 3.x–9.0, vSphere, vCenter, High Availability (HA), DRS, vMotion, Storage vMotion, EVC, vDS, NSX, vSAN
- **VMware Cloud Foundation 9:** SDDC Manager, VI workload domains, host commission, lifecycle automation, vSphere Lifecycle Manager (vLCM), 9.0-to-9.1 upgrade, Fleet Management, VCF Management Services, Identity Broker / SSO, brownfield import and inventory reconciliation
- **Operations & Analytics:** vRealize Operations 5/6/7, VCF Operations (HA / Continuous Availability, Cloud Proxy / VCF Operations Collector), VCF Operations for Logs, Log Insight, Skyline Advisor/Health; custom management packs (Management Pack Builder), super metrics, capacity reporting and forecasting, alerting
- **Storage & Cloud-Native:** vSAN ESA and OSA (disk groups, storage policies, capacity engineering); storage protocols — Fibre Channel (FC), NFS, VMFS, iSCSI, NVMe/TCP; Kubernetes via vSphere Kubernetes Service (VKS/TKG), CNS / First Class Disks and persistent volumes, NetApp ONTAP REST API, custom VCF Operations management packs and dashboards
- **Troubleshooting & RCA:** log analysis, root-cause identification, cluster-store / vpxd / NSX transport-plane / credential and certificate diagnostics; runbook and RCA authoring
- **Automation & Scripting:** PowerCLI, PowerShell, Python, Bash, Linux (Photon OS), SQL, VCF Suite / CASA and vim25 / CNS API integration, document automation
- **Infrastructure as Code & Cloud Automation:** VMware Aria Automation 8.18 (Cloud Assembly, cloud templates, Service Broker, secrets, cloud-init), Photon OS appliance deployment, REST API development (FastAPI / OpenAPI-Swagger, JWT auth), systemd / cron service automation
- **Cloud Platforms:** Microsoft Azure services, monitoring, security, cost optimization, Kusto
- **Disaster Recovery:** VMware Site Recovery Manager (SRM), EMC RecoverPoint, Zerto, Veeam, Hyper-V
- **Storage Platforms:** NetApp, EMC VNX, Dell EqualLogic, EMC Unisphere
- **Hardware Platforms:** Dell, HP, Cisco UCS, HP OneView
- **Performance Engineering:** capacity planning, optimization, baselining, diagnostics

PROFESSIONAL EXPERIENCE

Entelligence (Broadcom Premier Partner) -- Exelon Engagement (*Contracted via Virtual Control LLC*)

Senior VMware Cloud Foundation / VCF Operations Engineer — 2026 – Jul 2026

- Designed and delivered a **NetApp ONTAP to VCF Operations 9 monitoring integration** for a Fortune 100 energy utility (Exelon): built a custom **management pack** in Management Pack Builder that polls the **NetApp ONTAP REST API** to collect per-volume and per-aggregate capacity, performance (IOPS / latency / throughput), and storage-efficiency telemetry (49 metrics, 48 properties).
- Built the full analytics layer -- **single-pane dashboards**, list views, schedulable reports, **super metrics** (per-volume percent-free and efficiency ratio), and a complete **alert set** (6 alert types / 7 definitions: high latency, low free space, IOPS anomaly, aggregate over-commitment, efficiency drop, and not-collecting for volumes and aggregates).

- Packaged the solution as a client-facing **VMware by Broadcom High-Level Design** deliverable, with production deployment / build / config guides, per-team coordination hand-off packets, and a stakeholder Q&A brief.

Virtual Control LLC -- VCF 9 Engineering, Troubleshooting & R&D

Senior VMware Cloud Foundation Engineer (Independent Lab) — 2026 – Present

- Built and operate a **full nested VMware Cloud Foundation 9.0.1 environment** (vSphere, vSAN, NSX, SDDC Manager, VCF Operations, Aria Automation) end-to-end for capability development, testing, and client enablement.
- Integrated **VCF Single Sign-On with Active Directory** (embedded Identity Broker, AD/LDAP) and authored the end-to-end SSO setup runbook -- prerequisites, connection-security decision, and per-component role mapping.
- Authored verified operations references: a **VCF Operations Core Operations Runbook** with live-validated CASA and Suite API endpoints, health-check handbooks, and root-cause troubleshooting modules.
- Created the **VCF Health Check application** -- a comprehensive VCF 9 assessment platform (Bash engine plus Python) running 100+ automated checks across vCenter, NSX, SDDC Manager, vSAN, VCF Operations, Fleet / Lifecycle, and Aria Automation, producing graded HTML / PDF / JSON reports with per-component scoring, heatmaps, and historical trend analysis; delivered with a desktop GUI, a headless CLI engine, and a REST API (FastAPI / OpenAPI-Swagger).
- Automated the application's delivery with **VMware Aria Automation** (Infrastructure as Code) -- a Cloud Assembly cloud template that provisions it as a self-running Photon OS appliance with scheduled daily reporting and email delivery, hardened for reboot persistence via systemd and cron.
- **Recovered a brownfield SDDC Manager host-inventory desync** -- three ESXi hosts left orphaned (status None, no cluster reference) after an out-of-band vCenter redeploy -- by root-causing the aborting domain guardrails, remediating a stray host VMkernel tag, and driving the Sync Inventory (brownfield import) to completion, restoring all four hosts to **ASSIGNED with no vendor support case**.
- **Root-caused a vCenter vpxd service crash** to an ESX cluster-store (DKVS) reconfigure loop / half-joined replica (Broadcom KB 385346, resolved in vSphere 9.1); authored the RCA and stood up a **scheduled read-only monitor** that flags the escalation condition (alarm state / service recycle).
- Conducted a **fleet-wide appliance-credential audit and rotation** -- identified and remediated expired / expiring root passwords across VCF Operations, Fleet/LCM, NSX, and vCenter, disabled recurrence, and authored the root-cause analysis for the expiry pattern.
- **Validated and hardened fleet time services (NTP)** across all VCF components -- corrected NTP config drift, cleared the fleet-wide re-lock transient, and verified via the SoS NTP health check.
- Authored a **verified VCF 9.0-to-9.1 upgrade runbook and homelab feasibility assessment** from Broadcom's documented upgrade sequence -- prechecks, the new VCF Management Services deployment, License Server, and resource / HCL gating.
- Performed **CNS / First-Class-Disk hygiene analysis** -- cross-checked 99 flagged vSAN disk objects against the vStorageObject / CNS catalog, distinguishing live persistent volumes from reclaimable orphans and preventing deletion of in-use data.
- Recovered **SDDC Manager file-based backup** end-to-end after a cascading out-of-band redeploy -- rebuilt wiped NSX and vCenter service accounts and corrected stale inventory MoRefs, restoring all backup sub-tasks to SFTP.
- Resolved an **NSX transport-plane outage from a VDS-UUID mismatch** via NSX Manager redeploy and transport-node rebuild, restoring SDDC Manager, NSX, and vCenter integration.
- Migrated **VCF 9 from evaluation to fully licensed** via the disconnected subscription model (VCF cores plus vSAN), resolving the License-Management activation gap (Broadcom KB 409416).
- Performed **vSAN capacity engineering** -- reclaimed ~375 GB through orphaned-object and snapshot cleanup and expanded a disk group by 400 GB, clearing capacity alarms (74% to 60% utilization).
- Diagnosed and resolved a **cluster-wide vSAN write-performance collapse** by tracing a stray vSAN-tagged VMkernel adapter injecting a dead unicast peer, restoring ~10x throughput.
- Authored a **VCF 9 Platform API Handbook** and a library of runbooks and root-cause analyses spanning deployment, backup recovery, NSX, vSAN, SSO, credential lifecycle, and licensing.

Kyndryl / Citizens Bank / True Value / HCSC / NYMTA / Toyota

Mainframe Engineer -- Performance & Capacity Management — Dec 2019 – Feb 2026

- Engineered PowerShell automation to extract SQL metrics via ODBC, enabling rapid performance reporting and visualization.
- Built Excel-based dashboards with embedded SQL queries for real-time capacity insights.
- Delivered predictive capacity models improving resource planning and SLA compliance.

- Automated scheduled reporting (CSV / XLSX), reducing manual workload and improving operational efficiency.
- Led tuning initiatives reducing system overhead by 10% and MIPs consumption by 15%.
- Supported real-time incident investigations across mainframe and distributed systems.

Thielsch Engineering (Contracted via Virtual Control LLC -- VMware Partner)

Senior VMware Consultant — Mar 2016 – Dec 2025

- Fully redesigned and supported the vSphere environment over a 9-year engagement, upgrading through every major release from 5.5 to 8.0 U3G.
- Deployed VMware Cloud Foundation 9 in production and lab environments, including SDDC Manager, NSX, vSAN, ESXi 9.0, Fleet Management, VCF Operations, and VCF Operations for Logs.
- Built custom VCF Operations dashboards, alerting, and integrations for vCenter, NSX, and SDDC Manager. Configured VCF Operations for Logs for centralized syslog ingestion, enabling real-time log analysis and correlated troubleshooting.
- Configured and troubleshot HTTPS offline depot for VCF Fleet Management, including TLS certificate generation, Java trust store import via keytool, and vCenter Guest Operations API for remote appliance management.
- Developed a comprehensive VCF Health Check automation tool (75+ checks) with HTML reporting, heatmaps, SLA tracking, and historical trend analysis.
- Administered 150+ VMs across dual environments with high availability, and implemented 10 Gb trunking reducing latency by 30%.
- Led Azure migration, vCenter consolidation improving efficiency by 30%, and decommissioned EMC VNX arrays migrating workloads to NetApp storage.
- Designed and supported VMware SRM for resilient DR and seamless failover. Mentored team members and delivered strategic consulting across multiple VMware environments.

Advantasure / BCBS

VMware & Cloud Senior Architect — 2016 – 2019

- Delivered best-practice VMware design, VM hardening, and infrastructure modernization.
- Managed Azure services including Security Center, Monitor, Advisor, Change Tracking, and Reserved Instances.
- Executed vCenter consolidation reducing hardware footprint and support calls by 80%.
- Deployed and administered vROps 6.6 / 7.0, improving virtual server performance by 33%.
- Automated health checks with PowerShell, increasing operational efficiency by 27%.

Johnson & Johnson (VMware Partner)

VMware Senior Consultant / Technical Lead -- DEV-OPS NA — 2015 – 2016

- Managed hardware lifecycle and profile assignments using HP OneView.
- Administered 8,000+ VMs across 8 global vCenters and 400+ ESXi hosts.
- Implemented monitoring and troubleshooting workflows via vROps.
- Established performance baselines and optimization standards.
- Mentored engineering teams to elevate VMware SDDC capabilities.

Citizens Bank

Senior Infrastructure Engineer, AVP -- Virtualization Services — 2012 – 2015

- Co-founded and led the Virtualization Department.
- Configured HP SIM for 225 ESX/ESXi hosts using WBEM for real-time hardware monitoring.
- Served as Tier 3 VMware SME for enterprise operations.
- Rebuilt 130 ESX 4.0 hosts to 4.1 including firmware upgrades.
- Earned three Good Banking Awards and a Bank Credo Award.

ADDITIONAL EXPERIENCE

Hasbro Toys, EMC, VMware, Genzyme, McKesson -- Senior VMware and Infrastructure roles, 2000–2012.

CERTIFICATIONS

- VMware Cloud Foundation 9 Administrator

- VMware Certified Professional -- Data Center Virtualization (VCP-DCV), all versions 3 through 6.5
- AWS Cloud Practitioner
- AI Foundation Certificate (Kyndryl)
- AI Mainframe Modernization (Kyndryl)
- Microsoft Professional Orientation: Cloud Administration
- Microsoft IT Support: Cloud Fundamentals

EDUCATION & PROFESSIONAL DEVELOPMENT

Continuous professional development through VMware, Microsoft, AWS, and Kyndryl certification programs. 25+ years of hands-on engineering across enterprise virtualization, cloud platforms, and mainframe systems.